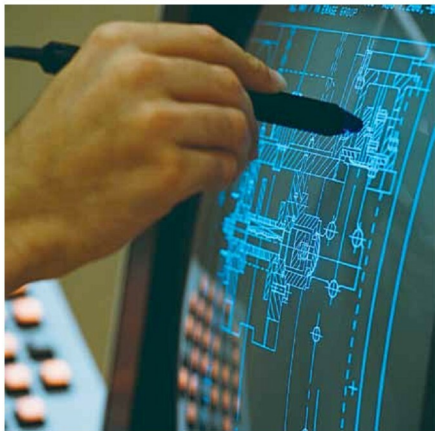


Frugal Engineering: The Mantra To Design Resource-efficient Embedded Systems and Chips

The article explores why it is critical to optimise resource consumption during the process of embedded systems and chips development, and the reasons that lead to resource drain. It also suggests a way forward to overcome these design hurdles



■ VANDANA SHARMA, WITH INPUTS FROM JANANI GOPALAKRISHNAN VIKRAM

"A small leak will sink a great ship."

— Benjamin Franklin,
American statesman

These words of Franklin indeed hold a lot of meaning and relevance for engineering teams

that design embedded systems and chips, as they well understand that resource-pilferage during the process of product design can lead to unwarranted outcomes. Resource-leakage has always remained one of the key challenges and primary concern of engineers designing chips and embedded systems.

Although every design engineer invariably tries to prevent resource-wastage, yet this is a phenomenon that seeps in during the process of product development. The reasons can be many, but the repercussions may range from increased power consumption and prolonged development cycle, to adverse effect on the system's eventual cost!

Thus, the embedded systems and chip designers aim to optimise the system's performance and keep the time-to-market as low as possible. From selecting the right processor for the onboard memory, to selection of other resources that can be populated on the system hence become critical choices. No wonder design engineers have begun to hail frugality in resource consumption as the key engineering goal.

Why is it so critical to be resource-efficient?

Before evaluating the reasons that make frugality in resource consumption a critical requirement, let's first understand what the term frugality means for an embedded system development engineer and chip designer.

"For an engineer designing embedded systems, achieving resource optimisation may include tasks like minimising the memory-footprint and the number of clock-cycles consumed. For a chip designer, frugality may mean minimising the area occupied on a die, or the amount of power consumed,"